GOOGLE CLOUD & GEMINI HACKATHON

NutriLens AR

Ambient Smart Glasses Nutrition & Macro Telemetry Agent

Zero-click nutrition logging. Look at any meal plate — Gemini Vision automatically segments complex regional dishes, projects micro-HUD macros onto your smart lens, and whispers real-time dietary coaching directly into your earpiece.

AI Engine: Google Gemini 3.8 Flash (Structured Outputs)

Deployment: Google Cloud Run Serverless Container

GitHub: github.com/deva96/nutrilens-ar

The Problem: Why Calorie Tracking Fails

Traditional nutrition apps create excessive friction, leading to 70%+ user abandonment.

1. Excessive Dining Friction

- Pulling out a phone at dinner breaks social conversations.
- Typing, searching, and estimating takes 3-5 minutes per meal.
- Users give up within 14 days due to logging fatigue.

2. Barcodes Fail Regional Meals

- Barcode scanners only work on packaged grocery goods.
- Multi-item regional platters (Thalis, Biryanis, Dosa, Curries) have no standardized labels.
- High inaccuracy on restaurant food.

3. Hidden Cooking Fats & Macros

- Traditional apps ignore hidden cooking oils, ghee, and sauces.
- Results in 400-600 kcal tracking errors every single day.
- Users miss fitness & health goals despite diligent logging.

The Solution: NutriLens AR Ambient Wearable

A hands-free smart glasses nutrition assistant that operates seamlessly in the background.

■ Zero-Click Gaze Capture

- Optical sensor automatically detects when gaze rests on food.
- Temple tap or blink triggers instant frame telemetry capture.
- Zero manual phone typing or scale weighing required.

■ Micro-HUD AR Projection

- Unobtrusive AR overlay projected directly onto smart glasses lens.
- Real-time breakdown: Calories, Protein, Carbs, and Fats.
- Live balance indicator vs. user's remaining daily budget.

■ Bone-Conduction Audio

- Ambient AI coach whispers a 2-second actionable portion tip.
- Personalized to user's daily goals (High-Protein, Deficit, Clean Bulk).
- Private, natural, and completely hands-free.

System Architecture on Google Cloud

High-performance multimodal pipeline combining edge simulation and serverless intelligence.

Layered Architecture Breakdown

Client / Optical Edge:
Smart Glasses Simulator + Web Speech Synthesis API

Vision Intelligence:
Gemini 3.8 Flash Multimodal Vision

Structured Outputs:
Pydantic Schema (deterministic JSON telemetry)

Serverless Compute:
Google Cloud Run / Streamlit Container

High-Availability:
Automated retry backoff & multi-model failover

Why Gemini 3.8 Flash & Google Cloud?

✓ Sub-2 Second End-to-End Latency
Fast enough for real-time smart glasses POV interaction.

✓ Complex Regional Food Segmentation
Accurately segments biryanis, curries, sushi, dosas, and sides.

✓ Strict Pydantic JSON Telemetry
Guarantees zero markdown hallucinations for HUD widgets.

✓ Serverless Scalability with Cloud Run
Scales from 0 to 100,000 requests with zero idle compute cost.

Live Demo, Validation & Next Steps

Ready for real-world optical wearables and everyday dining.

Prototype Validation Results

- **Zero-Click Latency**
  < 2.0s average response time
- **Segmentation Accuracy**
  Multi-dish regional recognition & hidden fat estimation
- **Audio Whisper Coaching**
  Natural conversational tone via Web Speech API
- **Reliability**
  Built-in automatic retry & capacity failover

Project Resources & Roadmap

Future Roadmap:

1. Native Android XR / Ray-Ban Meta glasses integration.
2. Real-time Continuous Glucose Monitor (CGM) sync.
3. Long-term dietary health coaching with Gemini Memory.

Submission Links:

- GitHub: https://github.com/deva96/nutrilens-ar
- Live Prototype: Deployed & Ready for Evaluation
- AI Engine: Google Gemini 3.8 Flash Multimodal